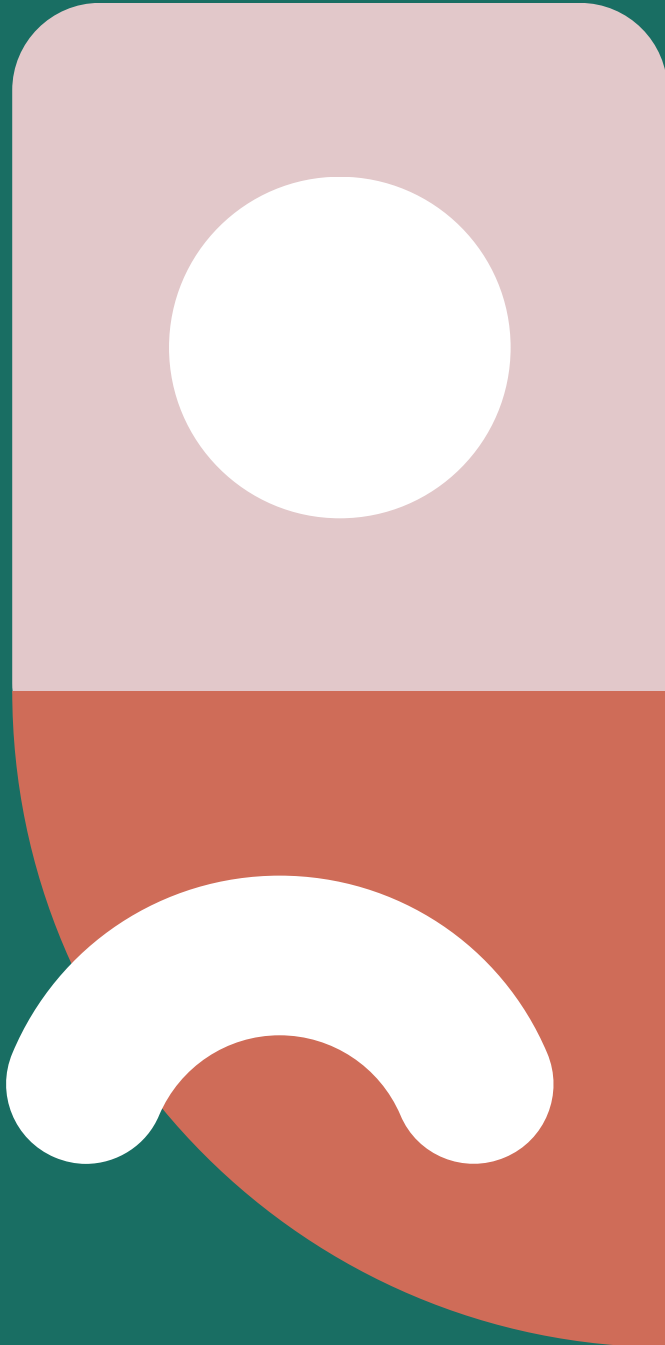
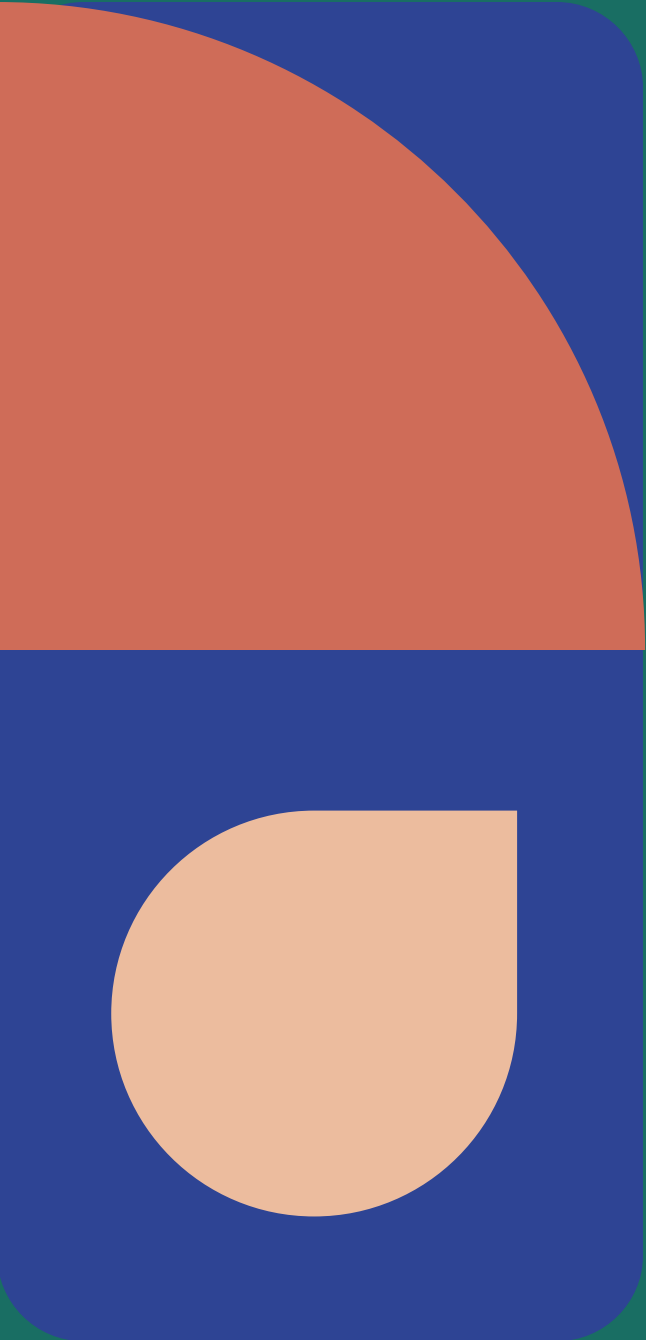


The logo graphic on the left consists of a light pink rounded square with a white circle inside, positioned above a red rounded square. The red square features a white, stylized, curved shape at its bottom left corner.

# PREPARE

Advancing Crop Price Prediction and Farmer  
Empowerment

The logo graphic on the right consists of a red rounded square positioned above a blue rounded square. The blue square features a white, stylized, curved shape at its bottom right corner, which is partially obscured by an orange circle.

# The Team

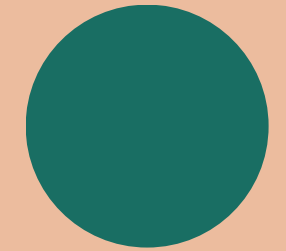
Under Prof.  
Mayank Ratan  
Bhardwaj

**Aryan Gosain**

**Kunal Ranjan**

**Rishit Anand**

**Varun SG**



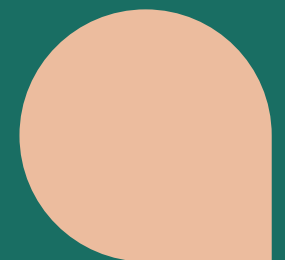
# Problems

## 1: Farmer's Price Blindness

Small and marginal farmers lack timely, accurate price forecasts and struggle to interpret market dynamics.

## 2: Ineffective Government Interventions

Governments and regulators lack reliable multi-day price prediction tools, making it difficult to set fair MSPs or plan timely market interventions.



# The Engine

- Input Data: Historical Price, Geographical Location, Weather
- Data Pre-processing: Outlier Removal, Normalization
- Modeling Core: Define and build upon the model chosen
- Output: Accurate Price Predictions

# The Chassis

- Prediction Service: Refined model is exposed via an API or server.
- Farmer Application: A mobile application for practical deployment.
- End-User Value: Empowers farmers with timely and accurate price forecasts

**Schematic**



# Deliverables

## **Make a New Model**

- Model Innovation
- Feature Engineering

## **Make an App-Based Interface**

Make a mobile app for the farmers.

## **Link the App to the Model**

Make APIs using FastAPI to get real time crop price prediction for the next 3/7 days (depends on how well our model performs).



# Sem 6

Extensions

## Explainable AI

Apply and Calculate SHAP values for input features. Adapt XAI methods to specifically explain the contribution of neighboring mandis

## Game Theory Based Market Dynamics

Construct a basic Non-Cooperative Game Theory Model involving two dominant 'players' (a large mandi vs. a collective of small farmers, or two competing mandis).



# Domain Research

## Data Sources

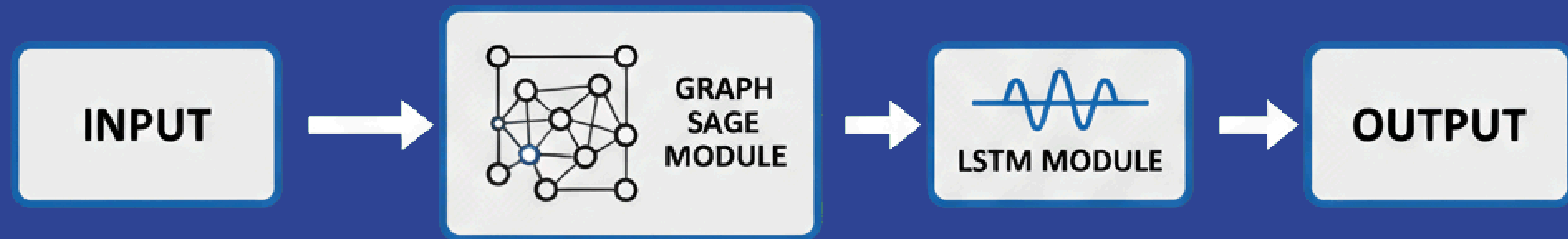
We analysed possible sources (AGM, eNam) for data and settled on the government Agmarknet portal.

<https://agmarknet.gov.in/home>

## Model and Methodology

We discussed different model options (such as GraphSage and other options), ways to impute the data (splining, mean values), and how existing apps look like in the agricultural domain.

# GNN-RNN Single Day Model



```
gnn_rnn_predictions.csv
1 timestamp,run_type,node_id,feature_idx,prediction,inference_time_ms,seq_len
2 2025-11-14T08:18:51.358020,single,0,0,884.5865478515625,72.49489997047931,7
3 2025-11-14T08:18:51.358020,single,1,0,1102.1534423828125,72.49489997047931,7
4 2025-11-14T08:18:51.358020,single,2,0,1396.380126953125,72.49489997047931,7
5 2025-11-14T08:18:51.358020,single,3,0,477.2625732421875,72.49489997047931,7
6 2025-11-14T08:18:51.358020,single,4,0,1152.1380615234375,72.49489997047931,7
7 2025-11-14T08:18:51.358020,single,5,0,991.510498046875,72.49489997047931,7
8 2025-11-14T08:18:51.358020,single,6,0,1027.0625,72.49489997047931,7,gnn_rnn
9 2025-11-14T08:18:51.358020,single,7,0,1150.300048828125,72.49489997047931,7,Cc
10 2025-11-14T08:18:51.358020,single,8,0,796.29931640625,72.49489997047931,7,g
```

**R2: 0.632725**

**This will be extended into a multi-day model now.**



# Experimentation

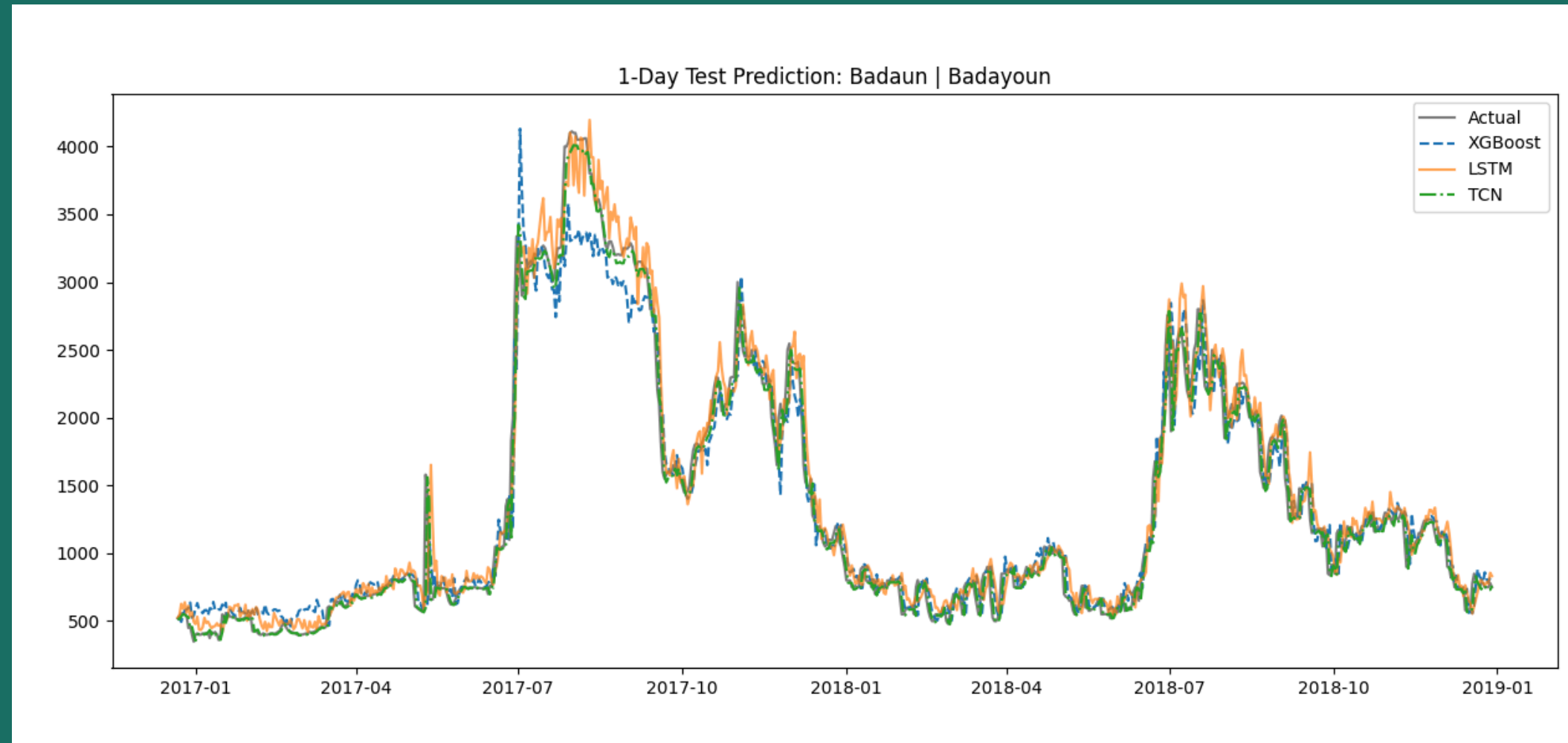
**We started off with only the mandis in UP to keep the goal achievable and focus on the scope of the app.**

## **Experimentation:**

- **Model architecture**
- **Model hyperparameters**
- **Data processing (normalization, etc.)**
- **App UI options**

train samples: 66,696  
test samples: 16,758

# 1 Day Predictions



$R^2 = 0.98$

train samples: 66,696  
test samples: 16,758

 Metrics

| Period               | Model Variant | RMSE   | R <sup>2</sup> |
|----------------------|---------------|--------|----------------|
| 1 <sup>st</sup> Day  | XGBoost       | 232.64 | 0.94           |
|                      | LSTM          | 189.51 | 0.96           |
|                      | GAT-LSTM      | 421.83 | 0.68           |
|                      | TCN           | 117.96 | 0.98           |
| 15 <sup>th</sup> Day | XGBoost       | 537.36 | 0.69           |
|                      | LSTM          | 660.02 | 0.54           |
|                      | GAT-LSTM      | 804.10 | 0.34           |
|                      | TCN           | 460.96 | 0.77           |

can still be improved,  
however this was more of a test to see what worked

**Extended to all APM mandis across India**

**21 → 279 (Tomato)**

**57 → 337 (Wheat)**

**This is where GAT-LSTM took the lead over other existing models.**

# GAT-LSTM

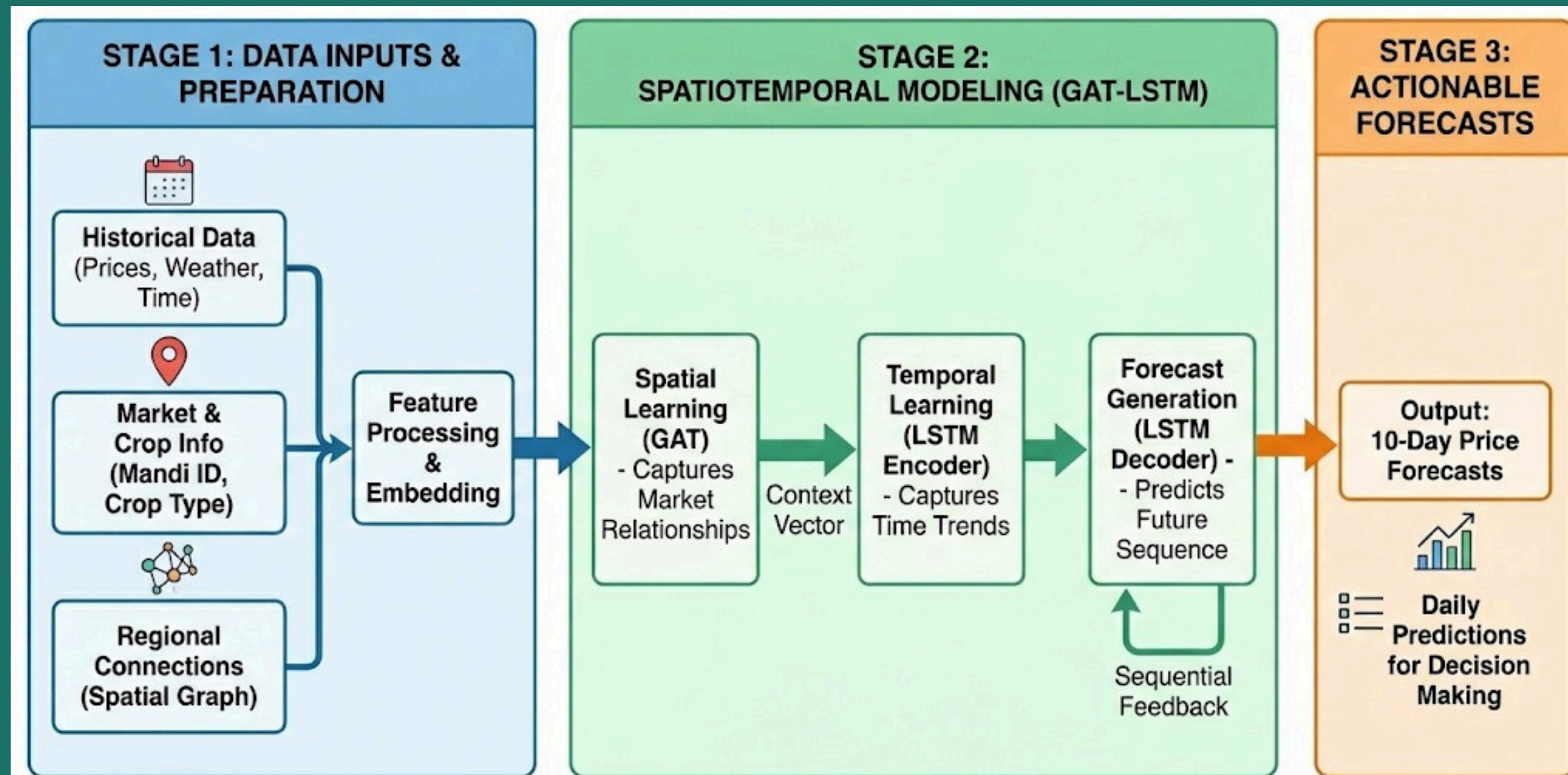


Figure 1: Application Flow of the GAT-LSTM Price Forecasting System



train samples: 736,093  
test samples: 92,012  
val samples: 92,012

Metrics

| Day | RMSE (₹/100kg) | MAE (₹/100kg) |
|-----|----------------|---------------|
| 1   | 252.59         | 177.53        |
| 2   | 288.60         | 198.90        |
| 3   | 314.08         | 217.34        |
| 4   | 333.61         | 231.02        |
| 5   | 348.95         | 241.72        |
| 6   | 361.51         | 250.93        |
| 7   | 371.87         | 258.71        |
| 8   | 381.00         | 265.79        |
| 9   | 388.82         | 272.09        |
| 10  | 395.71         | 277.85        |

R<sup>2</sup> = 0.80

train samples: 919,327  
test samples: 114,916  
val samples: 114,916



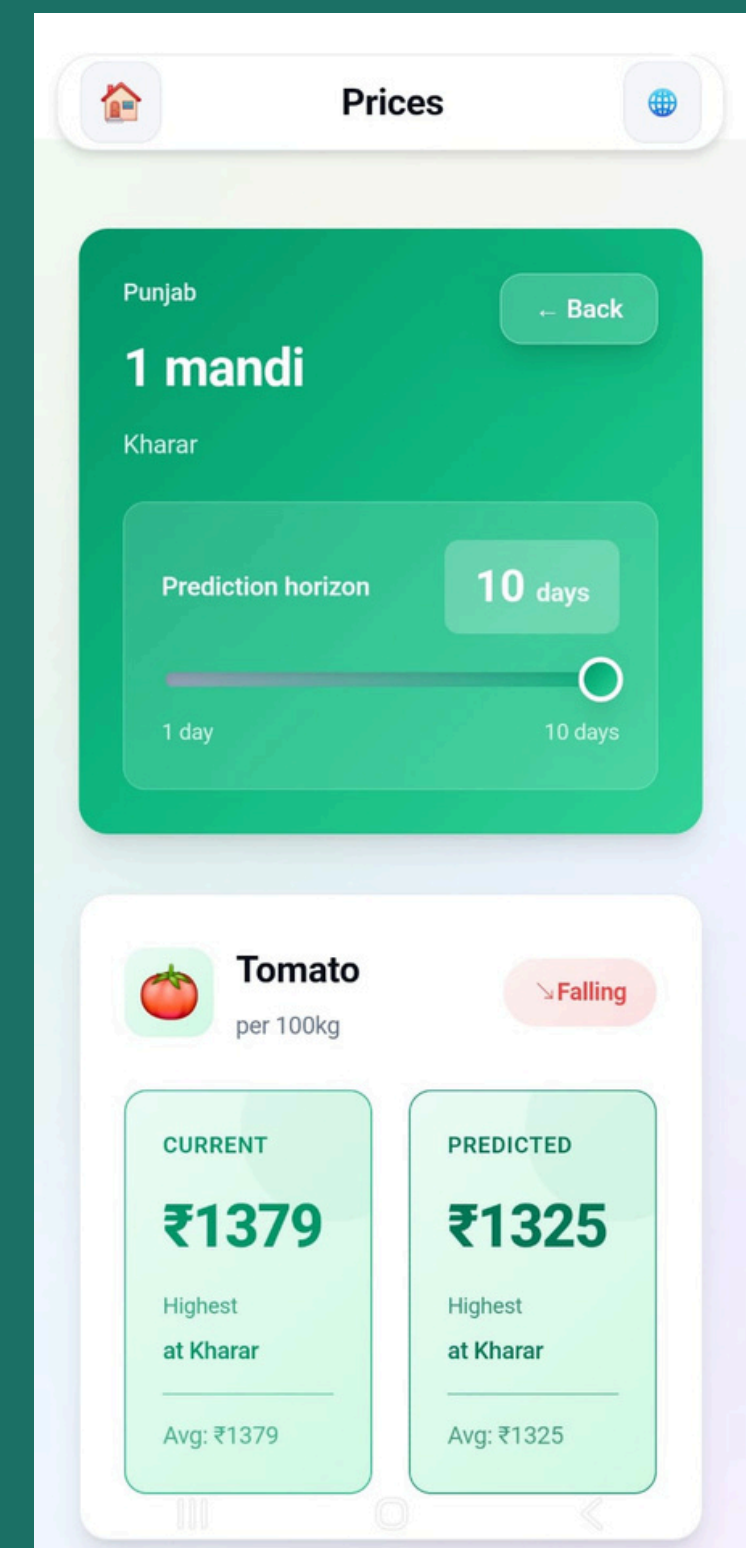
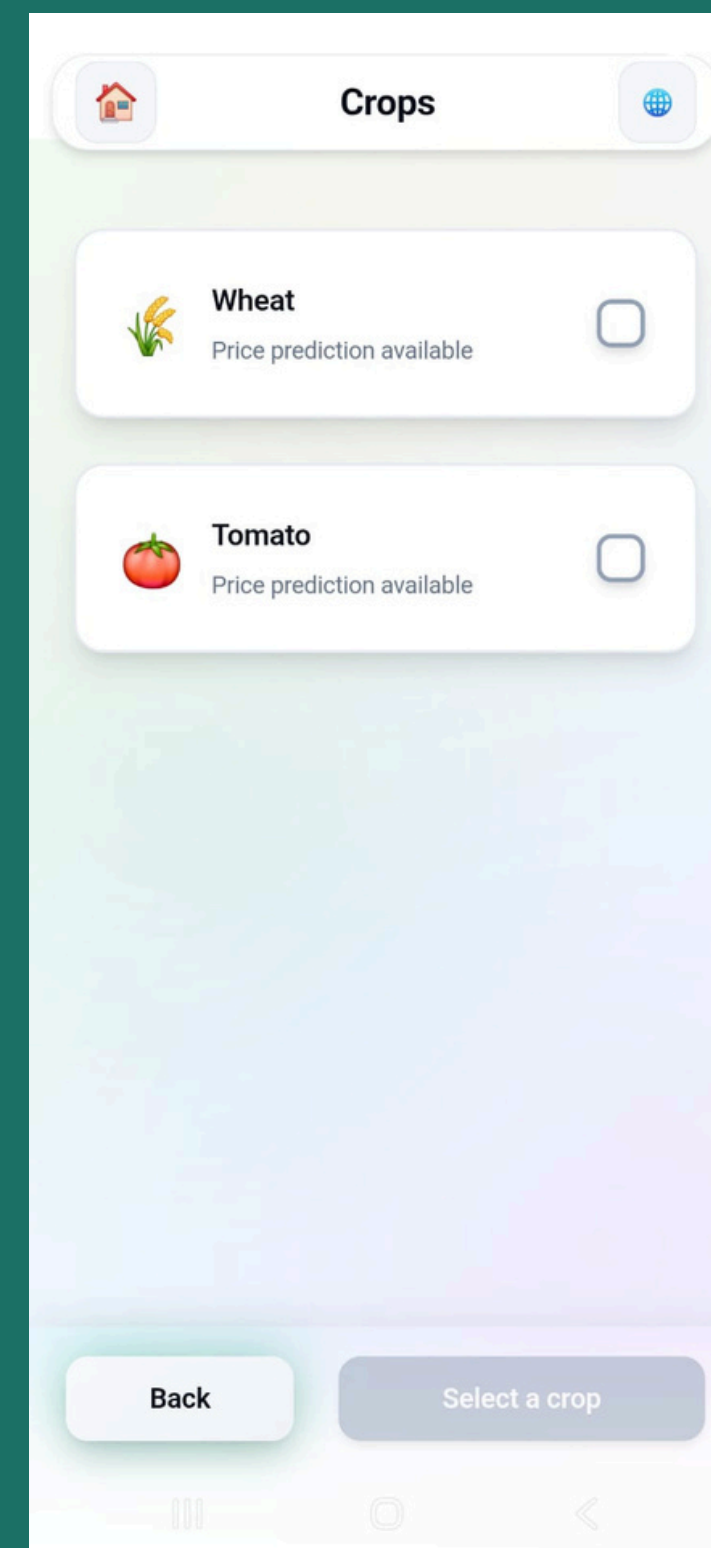
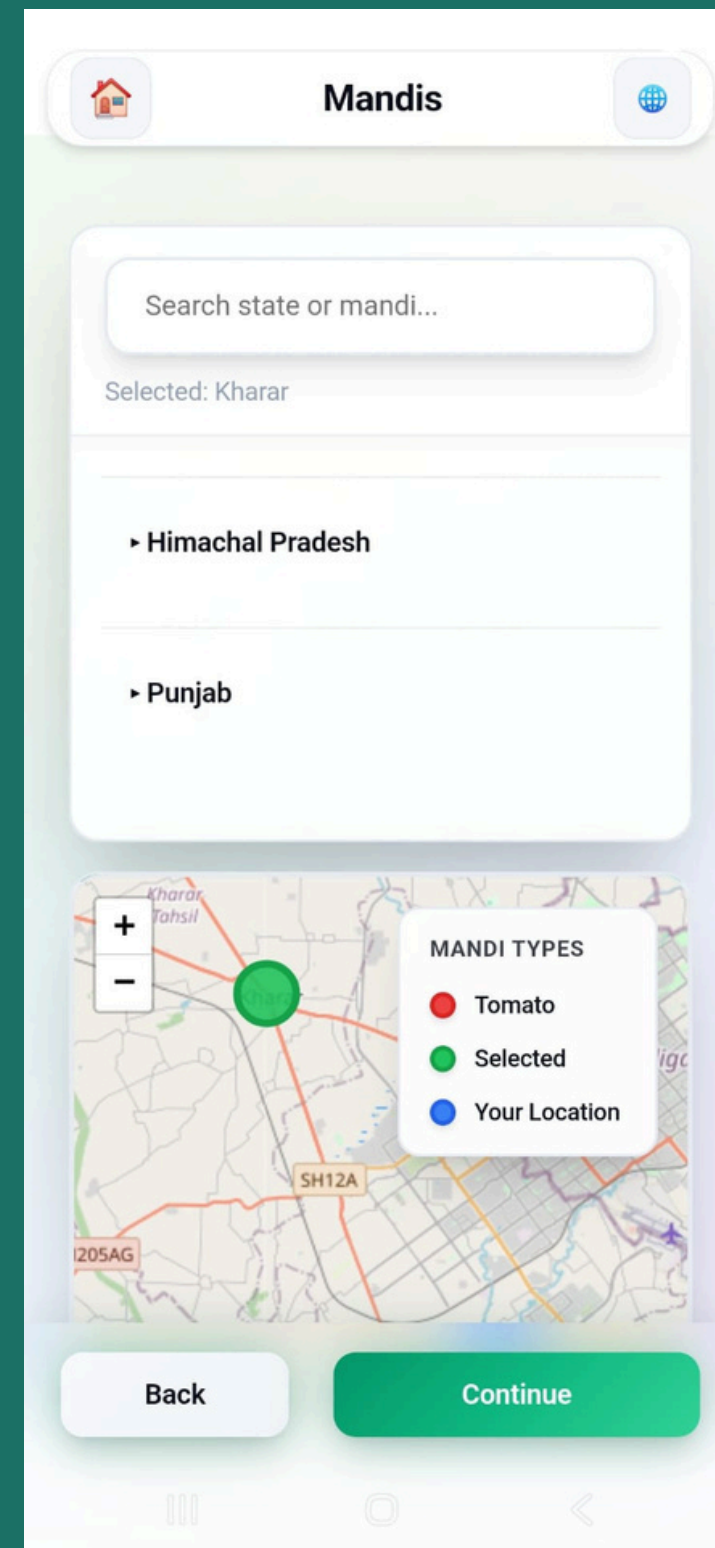
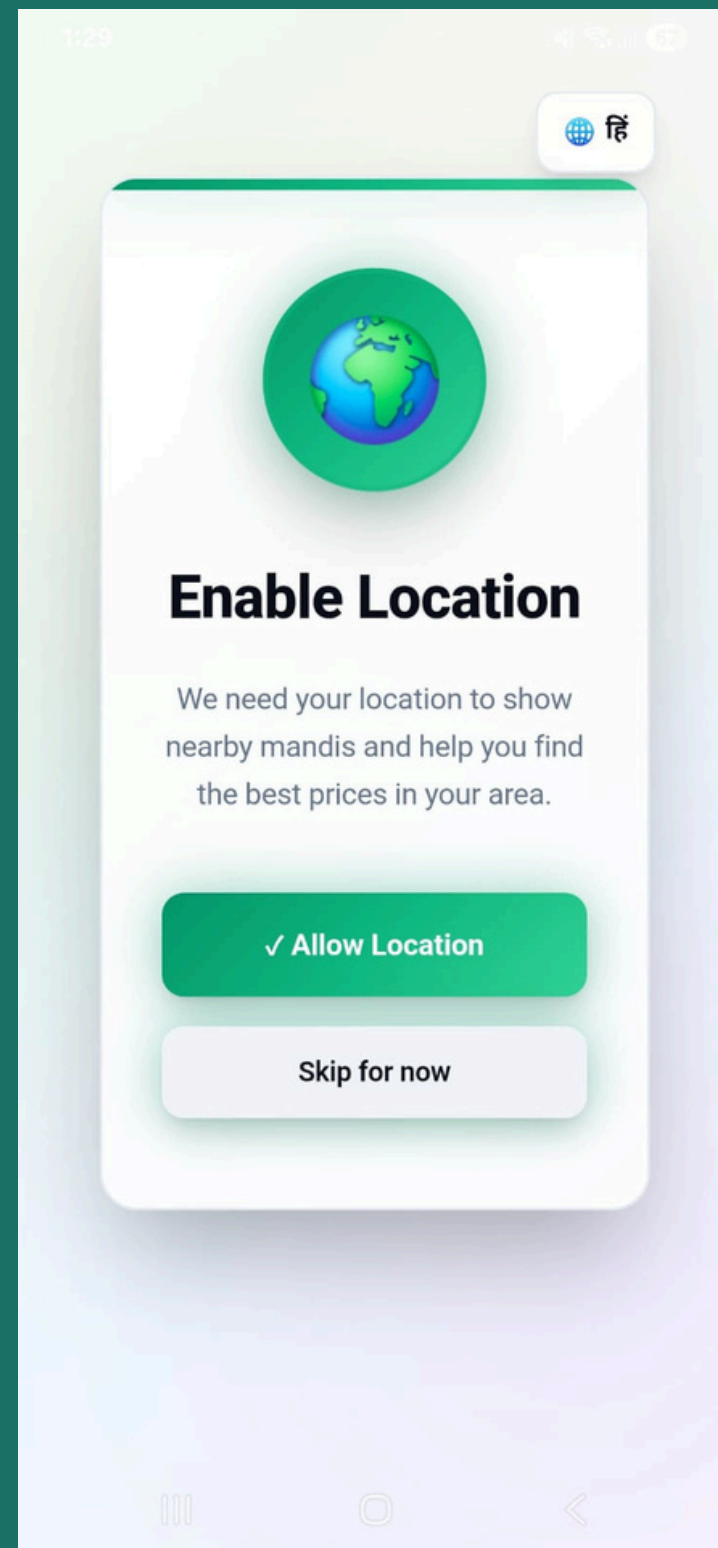
Metrics

| Day | RMSE (₹/100kg) | MAE (₹/100kg) |
|-----|----------------|---------------|
| 1   | 146.51         | 76.00         |
| 2   | 158.40         | 75.70         |
| 3   | 167.15         | 79.79         |
| 4   | 174.33         | 81.23         |
| 5   | 179.95         | 82.08         |
| 6   | 183.63         | 82.74         |
| 7   | 185.85         | 83.39         |
| 8   | 187.44         | 84.21         |
| 9   | 188.95         | 85.29         |
| 10  | 190.45         | 86.53         |

R<sup>2</sup> = 0.67



# Demo







## Novelty

- Capturing the interdependent nature of different markets for food pricing.
- There are **no current consumer facing solutions** that provide price predictions.

## Real World Application

This helps farmers to decide when to harvest crops and where to sell those crops for optimal returns. The format of the application also makes it accessible for farmers and does not have a significant learning curve.

# Deliverables Achieved



## **Make a New Model**

- Model Innovation
- Feature Engineering

## **Make an App-Based Interface**

Make a mobile app for the farmers.

## **Link the App to the Model**

Make APIs using FastAPI to get real time crop price prediction for the next 3/7 days (depends on how well our model performs).



# Extensions

**Improve the multi-day model accuracy.**

**Incorporate a wider variety of crops (seasonal + cash).**

**Improve query architecture.**

API → caching results.

# Sem 6

Extensions

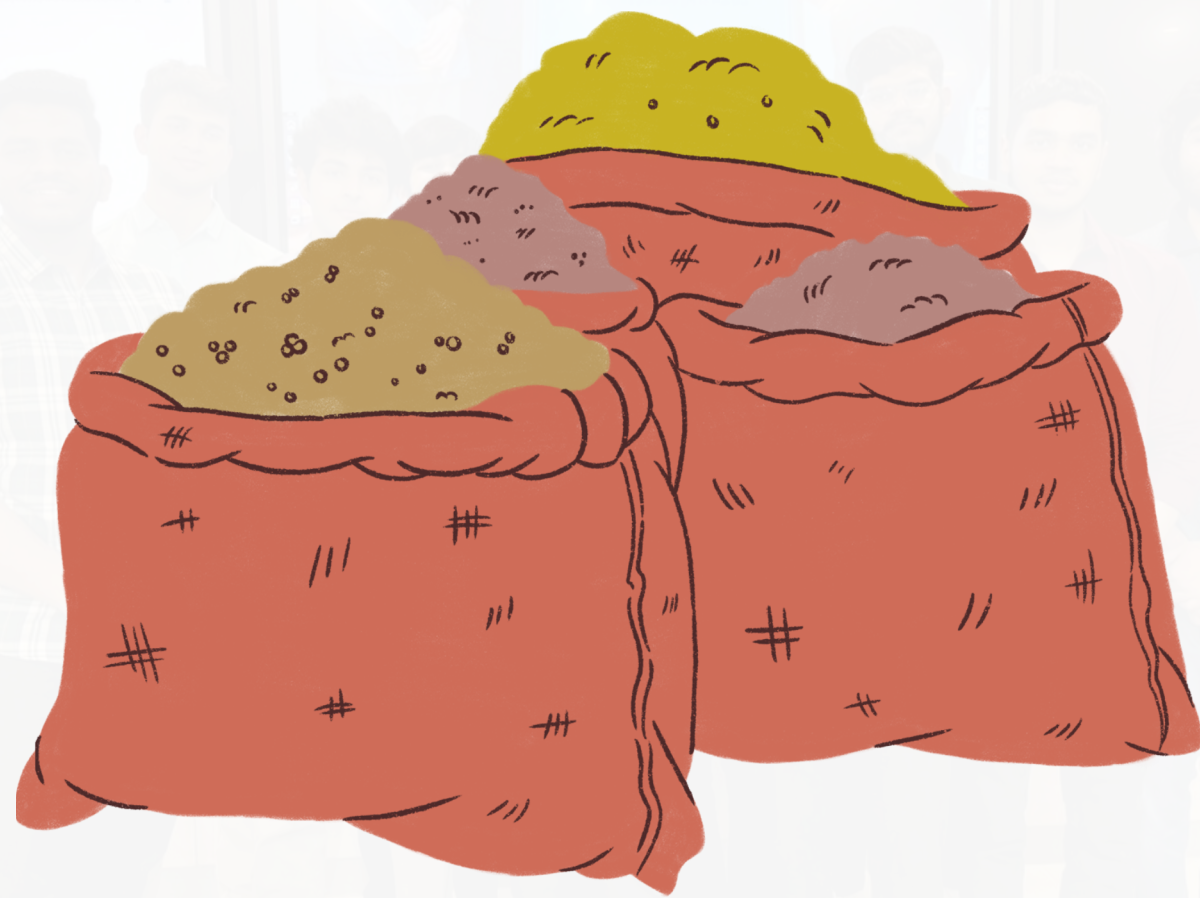
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thank



you